MILITARY SPECIFICATION

PUMPING UNIT, HYDRAULIC AF/M27M-1 AND AF/M27M-2

1. SCOPE

1.1 This specification covers two types of four-wheel-mounted, gasoline-engine-driven, hydraulic pumping units, designated AF/M27M-1 and AF/M27M-2 (see 6.2).

1.1.1 The AF/M27M-1 designation covers a winterized pumping unit.

1.1.2 The AF/M27M-2 designation covers a nonwinterized pumping unit.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

GG-G-76	Gages, Pressure And Vacuum, Dial Indicating (For Air, Steam, Oil, Water, *Ammonia And Chloro-Fluoro Hydrocarbon* Gases)
TT-N-95	Naphtha, Aliphatic

Military

MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-H-4607	Heater, Engine And Shelter, Ground, Type H-1
MIL-H-5440	Hydraulic Systems, Aircraft, Types I And II, Design, Installation, And Data Requirements For
MIL-R-5520	Reservoirs, Aircraft Hydraulic, Non-Separated Type
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, And Ordnance
MIL-M-8090	Mobility Requirements, Ground Support Equipment, General Requirements For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-B-11188	Battery, Storage, Lead-Acid
MIL-E-11275	Engine, Gasoline, Industrial Type, General Specification For

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STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers And Trailer Dollies; Preservation And Packaging Of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For
MIL-STD-808	Finishes, Protective And Codes, For Finishing Schemes For Ground And Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MS27766	Shroud For Aerospace Ground Equipment
MS28759	Hose Assembly, Rubber Hydraulic And Pneumatic, (3,000 PSI), Flared Tube
MS33514	Fitting End, Standard Dimensions For Flareless Tube Connection And Gasket Seal
MS33515	Fitting End, Standard Dimensions For Bulkhead Flareless Tube Connections
MS33649	Bosses, Fluid Connection Internal Straight Thread
MS33656	Fitting End, Standard Dimensions For Flared Tube Connection And Gasket Seal

Air Force-Navy Aeronautical

AN816	Nipple-Flared Tube And Pipe Thread
AN824	Tee-Flared Tube
AN827	Cross-Flared Tube
AN834	Tee-Bulkhead And Universal Flared Tube
AN924	Nut Plain Hexagon, Tube Bulkhead And Universal Fitting
AN929	Cap Assembly, Pressure Seal Flared Tube Fitting
AN6236	Filter Element-Hydraulic Replaceable Micronic Reservoir Type
AN6249	Valve, 3000 PSI Hydraulic Check

DRAWINGS

Ordnance Corps

7041347	Lunette Eye

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Air Force

43A13905	Connector Assembly-Hydraulic Hose, Female
43A13906	Connector Assembly-Hydraulic Hose, Male
48B7385	Ring-Cargo Tie Down
48B7796	Ring Assembly-Tie Down, 10,000 Pound
48B6564	Ring-Eyebolt, 25,000 Pound Cargo Tiedown
50C24033	Coupling-Male, 21 Inch Air Duct

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMEMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Components. The pumping unit shall consist of the following major components.

	Description	See Requirement
a.	Hydraulic system	3.9
b.	Gasoline engine and accessories	3.10
c.	Trailer, control panel, and housing	3.11

3.3 General. The general requirements for metals, fluid velocity limitations, removal of entrapped air, fluids, filters, and cleaning of parts and systems specified in MIL-H-5440 apply as requirements of this specification. Where the requirements of MIL-H-5440 and this specification conflict, this specification shall govern.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as specified in 3.4.1 and 3.4.2.

3.4.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS and AN) without alteration

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.4.2 Standard parts. With the exception specified in 3.4.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

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3.5 Materials.

3.5.1 Protective treatment. When materials are used in the construction of the pumping unit that are subject to deterioration when exposed to climatic and enviornmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

* 3.5.2 Fittings. All fittings and threaded bosses in the hydraulic system shall be as specified in MS33514, MS33515, MS33649 or MS33656. Use of tapered pipe threads shall be avoided. Use of pipe threads requires approval from the procuring activity. All 3,000 pounds per square inch pressure fittings shall be of the steel and bulkhead type. High-pressure fittings shall use a locknut with a backup gasket.

3.6 Design. The pumping unit shall be designed to be a self-contained, mobile pumping unit providing not less than 4 gallons per minute (gal/min) total flow at 3,000 pounds per square inch gage continuously with hydraulic components arranged as shown on Figure 1. Components shall be arranged for maximum ease of operation and maintenance consistent with the functional requirements specified herein.

3.6.1 The pumping unit shall be designed for continuous-duty operation from sea level up to and including 6,000 foot (ft) pressure altitude.

3.7 Construction. The pumping unit shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.7.1 Maintenance. The pumping unit shall be constructed so that adjustments, repairs, and overhaul can be easily accomplished by the personnel of operating units and overhaul bases. The pumping unit shall be constructed so that simple handtools may be used for maintenance and servicing and no special tools will be required.

3.7.2 Air transportability. The pumping unit structure, components, trailer, and mountings shall be constructed to withstand the acceleration loadings specified in MIL-A-8421.

3.8 Performance. The pumping unit shall be capable of withstanding the following conditions without detrimental effect to subsequent operation:

a. Operating temperatures ranging from -20° to +130° Fahrenheit (F) and storage in temperatures ranging from -65° to +160°F. The AF/M27M-1 pumping unit shall be capable of operating in temperatures as low as -65°F when the winterization kit specified in 3.12 is installed.

b. A relative humidity of 95 to 100 percent at an operating temperature of 130°F and a storage temperature of 160°F.

c. Exposure to salt-sea atmosphere.

d. Rainfall as encountered in any locale.

e. Fungus growth as encountered in tropical climates.

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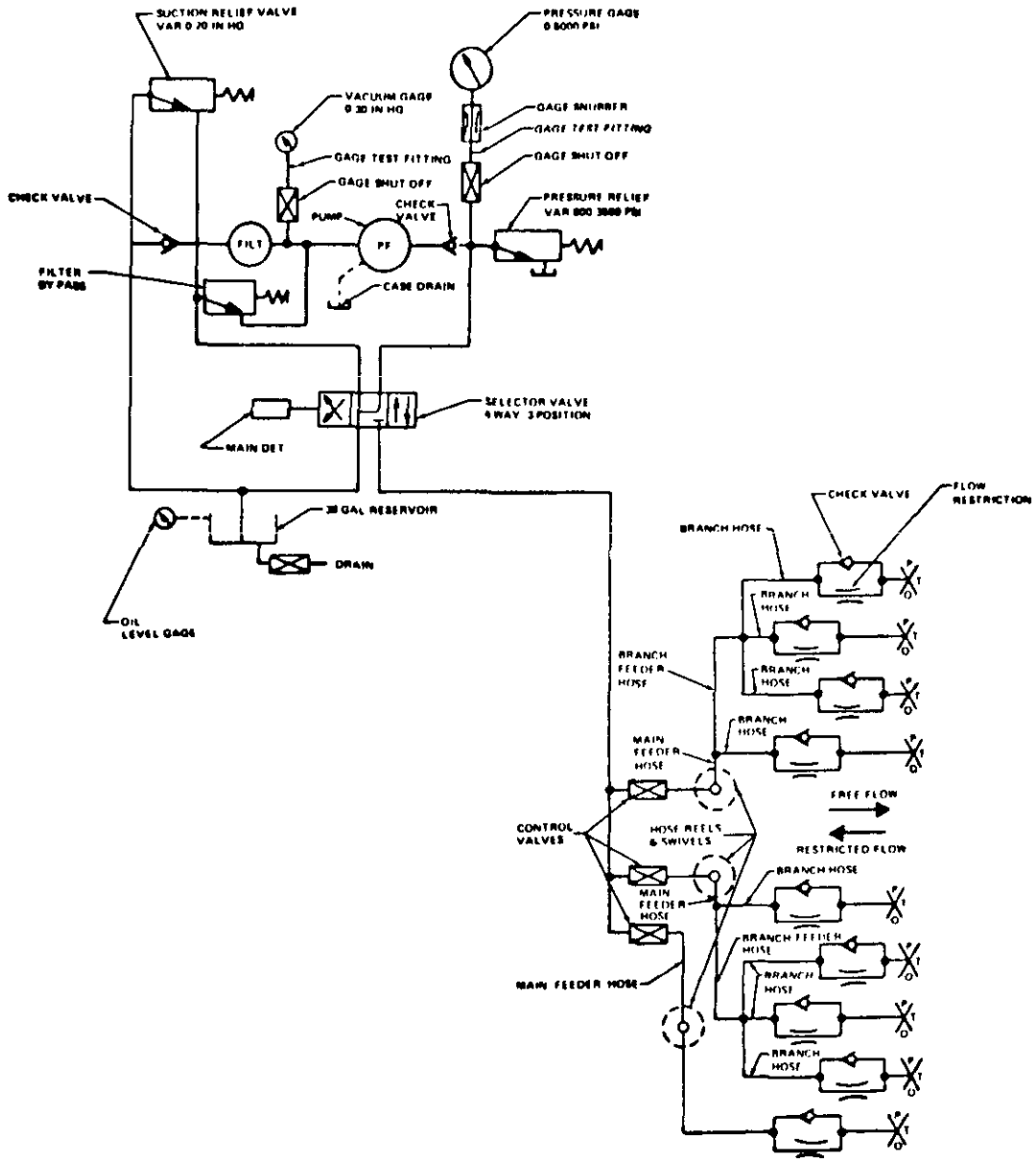


FIGURE 1. HYDRAULIC SYSTEM

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3.8.1 Operating positions. The pumping unit shall be capable of functioning satisfactorily when its normal operating plane is at an angle of 8-1/2 degrees in any direction from horizontal.

3.8.2 Noise. The noise level of the pumping unit shall not exceed 108 decibel (dB) measured at 10 ft in any direction from the pumping unit at any frequency within the audible range.

* 3.8.3 Electromagnetic interference. The pumping unit shall be suppressed for electromagnetic interference when tested to the requirements of MIL-STD-461, Class IIIC, 0.15 through 400 megahertz (MHz) (radiated).

3.9 Hydraulic system.

3.9.1 Hydraulic pump. The hydraulic pump shall be a fixed-displacement, axial, piston-type pump of adequate rating to meet the requirements specified herein.

3.9.1.1 Pump case drain. A pump case drain incorporating a relief valve, if required, shall be provided to maintain pump case pressure at a value recommended by the pump manufacturer, and return leakage to the reservoir.

3.9.2 Pressure check valve. A pressure check valve, similar to an AN6249-8 valve modified for 3,500 pounds per square inch, shall be provided as shown on Figure 1 to prevent reverse flow through the pump.

3.9.3 High-pressure gage. A flush-mounted, corrosion-resistant, high-pressure gage having a 4-1/2 inch diameter dial and a steel-alloy bourdon tube shall be provided. The high-pressure gage shall be accurate to within 2 percent, or less, of full-scale and shall conform to GG-G-76, except where conflicts exist, this specification shall govern. The gage shall provide a range of from 0 to 5,000 pounds per square inch and shall incorporate a means to restore the accuracy of the gage under all service conditions. It shall be suitable for operation at from -65° to +130°F. The gage shall be of the recess type provided with a mounting flange at the front of the gage. The mounting flange and dial shall not extend more than 5/8 inch beyond the external surface of the panel. Good-quality, stainless-steel movements shall be provided on all gages. Gages subjected to continuous surges and extreme pressure fluctuations shall be protected by a hydraulic dampener.

3.9.3.1 Mounting. Gages shall be mounted by screws and tapped panel holes to permit removal by one operator from in front of the panel.

3.9.3.2 Connection. Gages shall be connected to the panel with flexible AN- or MS-type hose, or commercial equivalent, of sufficient length to permit removal and installation from the front of the panel.

3.9.3.3 Gage lines. The high-pressure gage line shall be provided with a needle shutoff valve, an accumulator-type pulsation dampener, and gage test fitting. The vacuum gage line shall be provided with a needle shutoff valve and a gage test fitting.

3.9.3.3.1 Shutoff valves. The needle shutoff valves shall be of the panel-mounted type with a corrosion-resistant steel stem. The valves shall have the manufacturer's cold-working pressure rating stamped on the valve body with an arrow indicating the direction of flow. The valves shall be suitable for working pressures up to 5,000 pounds per square inch.

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* 3.9.3.3.2 Gage test fittings. The following fittings shall be mounted on the control panel as shown on Figure 1, adjacent to the appropriate gage, and the panel marked to indicate that the fittings are gage test connections:

a. An AN834-4 tee

b. An AN924-4 nut

c. An AN929-4 cap

* 3.9.4 High pressure relief valve. A high pressure relief valve (see Figure 1) adjustable from 800 to 3,500 pounds per square inch gage without the use of tools, and accessible at the front of the control panel, shall be provided.

3.9.5 Selector valve. A four-way, three-position selector valve designed for 3,000 pounds per square inch working pressure shall be provided as shown on Figure 1. The selector valve shall utilize O-rings to prevent external leakage. The selector valve shall direct pressure to the hoses and draw fluid from the reservoir while in the first position, dump pressure into the line from the pump to the reservoir while in the second position, and draw fluid from the hoses and dump fluid into the reservoir while in the third position. The valve handle position shall be marked up, hold, and down and shall correspond to the direction the jacks will be traveling.

* 3.9.6 Flow control valves. Three 3/8 inch or larger needle valves shall be provided to control the flow to and from the three hose reels specified in 3.9.7. The valves shall be suitable for operating pressures up to 3,500 pounds per square inch and for controlling flow within ±1/4 gal/min over the range of 0 to 4 gal/min.

3.9.7 Hose reels. Three hose reels shall be provided and mounted within the outline of the housing, one hose reel for each complete hose set specified in 3.9.13. The hose reels shall be mounted at the same opening of the housing so that the three hose sets may be pulled out through the same end of the housing. Fluid flow to and from the control valves shall enter its respective hose reel through swivel fittings and thence connected to the hose sets.

3.9.7.1 Handcrank. A handcrank shall be provided to turn the hose reels. The handcrank shall extend to the outside of the enclosure and shall be removable. Fasteners shall be provided in the housing for storage of the handcrank when not in use. The hose reel shall turn one revolution for each two revolutions of the handcrank and shall be connected by means of gears or a metal-chain drive.

3.9.8 Reservoir. An aluminum or stainless-steel, hydraulic-fluid reservoir shall be provided and mounted within the housing. The reservoir shall have adequate capacity to supply at least 38 gallons (gal) of fluid to the system when inclined 8-1/2 degrees from vertical, in any direction, plus additional space for a 5-gal expansion above the normal full level, and at least a 2-gal space below the outlet to the system for the collection of moisture and impurities.

3.9.8.1 The reservoir shall conform to the requirements for obstruction to flow, structural design, shell, seams, structural strength, baffles, fluid connections, sump, drain, vent, vent filter, vent line, return line inlet,

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filler unit, finish, and filling time of MIL-R-5520. Clearance shall be provided around the filler opening to allow rapid filling from 1-gal cans while the pumping unit is in operation.

3.9.8.2 Access plates. One or more access plates, at least 6-by-6 inch, shall be provided to allow visual inspection and manual cleaning of the interior of the reservoir.

3.9.8.3 Drain valve. A drain valve shall be provided for completely draining the reservoir. Clearance shall be provided around the drain outlet to permit rapid drainage into 1-gal containers.

3.9.9 Fluid level indicator. A fluid level indicator shall be provided to indicate the quantity of available fluid in the reservoir. The indicator shall be accurate to within ±2 gal of actual quantity. The indicator shall be clearly visible to the operator standing at the control panel.

3.9.10 Low-pressure filter. A suction line filter shall be provided and located as shown on Figure 1. The filter shall utilize a minimum of one replaceable AN6236-3 filter element or two replaceable AN6236-2 filter elements. Filling and draining plugs and air bleed valves shall be furnished and clearance allowed around each item for draining, bleeding, filling, and replacing the filter elements.

3.9.10.1 Bypass relief valve. The low-pressure filter shall be protected with a bypass relief valve adjusted to prevent collapse of the filter element due to excessive pressure drop across the filter.

3.9.11 Low-pressure gage. A flush-mounted, low-pressure gage with a 3-in, or larger, diameter dial shall be furnished. The gage shall be accurate to within 2 percent or less of full-scale range and shall provide a range of 0 to 30 in mercury vacuum. It shall incorporate a manual adjustment to restore the zero of the gage in the field under all service conditions.

3.9.11.1 Check valve. A preset check valve or relief valve shall be provided as shown on Figure 1. The valve shall prevent static or surge pressures in the low-pressure lines from exceeding 20 pounds per square inch gage.

* 3.9.12 Suction relief valve An adjustable relief valve that can be manually set to operate at any pressure from 0 to 20 in mercury vacuum shall be provided as shown on Figure 1 to prevent starving and cavitation of the pump if excessive vacuum is developed when lowering the aircraft jacks, and the aircraft weight is off the jacks.

* 3.9.13 Hose sets. The following hoses and fittings shall be furnished and connected in the sequence listed to make up the three hose sets as shown on Figure 1. The main feeder hose for each hose set shall then be connected to its respective hose reel.

a. Two MS28759-8 main feeder hose assemblies, 30 ft long.

b. Two AN824-8 tees.

c. Two MS28759-8 branch feeder hose assemblies, 25 ft long.

d. Two AN827-8 tees.

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e. Two MS28759-8 branch hose assemblies, 25 ft long.

f. Six MS28759-8 branch hose assemblies (connected to tees and crosses specified in b and d) 10 ft long.

g. Nine AN816-8 nipples

h. Nine automatic flow regulators to provide a directional flow control to allow unrestricted flow to the jacks and to restrict return flow from the jacks to 0.7 ±0.05 gal/min at 3,000 pounds per square inch gage.

i. Nine connector assemblies in accordance with Drawing 43A13905.

j. One MS28759-8 main feeder hose, 90 ft long, fabricated from not more than two pieces of hose with the connector conforming as near as possible to the curvature of the adjacent hose on the reel when the hose is stored on the reel.

3.10 Gasoline engine and accessories.

3.10.1 Gasoline engine. A four-cycle gasoline engine in accordance with MIL-E-11275, group II, shall be provided. The gasoline engine shall be of the four-cylinder type.

3.10.2 Gasoline engine accessories. In addition to other items as may be necessary to meet the requirements specified herein or in MIL-E-11275, the following items shall be provided. In the event of conflict between MIL-E-11275 and this specification, this specification shall govern.

3.10.2.1 The engine shall be provided with a positive displacement-type fuel pump, fuel filter with glass collecting bowl, oil filter, cooling system, and an oil-bath-type carburetor air intake filter.

3.10.2.2 Exhaust system. An exhaust system shall be furnished and shall include an exhaust collector, muffler, and spark arrester.

3.10.2.3 A constant-speed, adjustable range governor shall be provided to maintain the selected speed to within 5 percent of governor setting under all load conditions. The range shall be adjustable by hand at the control panel and shall provide for a range of speed settings from the minimum to the maximum operating speeds recommended by the engine manufacturer. A manual override shall be provided to adjust the throttle to the idling speed recommended by the engine manufacturer. A manual choke shall be provided to aid in starting the engine.

* 3.10.2.4 An electrical starting system, manual starting system, and battery-charging system with relay shall be provided. The electrical system shall be 12 volts.

3.10.2.5 Battery. A battery conforming to MIL-B-11188 capable of starting the pumping unit at -20°F without application of external heat to any part of the pumping unit shall be provided. The battery shall be securely mounted within the enclosure and shall be readily accessible for removal or servicing.

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3.10.2.6 Fuses or circuit breakers shall be installed as required to protect the electrical system and wiring against faults and overload currents.

3.10.2.7 Fuel tank. A fuel tank of adequate capacity to provide fuel for at least 4 hours of full-load operation shall be provided and mounted within the outline of the pumping unit housing. The fuel tank and supporting frame shall be sufficiently rigid to prevent damage due to vibration and handling.

3.10.2.7.1 Filler opening. A filler opening of not less than 2 inch inside diameter shall be provided and located to facilitate filling from 5-gal containers. The filler opening shall be provided with a screen for removing impurities.

3.10.2.7.2 Cap. The fuel tank cap shall be painted red. The fuel tank cap shall be securely attached to the fuel tank by means of a metal chain.

3.10.2.7.3 Sump. A sump shall be provided in the fuel tank for collection and accumulation of not less than 1 pint of water. A drain shall be provided and located at the bottom of the sump to permit complete drainage of the fuel tank. Fittings shall be installed to carry the drainage from the fuel tank to the outside of the housing.

3.10.3 The engine shall be provided with a starter control, tachometer, hourmeter, oil pressure gage, temperature indicator, ammeter, and such other instruments and controls as may be required.

3.11 Trailer.

3.11.1 Trailer, control panel, and housing. The components of the pumping unit shall be mounted and assembled on a trailer and covered by a weather-resistant housing. The pumping unit shall meet the requirements of MIL-M-8090 for type II, group C, mobility.

3.11.1.1 Frame. The trailer frame shall be of welded-steel construction. It shall be from 36 to 40 inches wide and from 48 to 66 inches long. The ratio of width divided by length shall not exceed three-fourths.

3.11.1.2 Parking brakes. The trailer shall be provided with two-wheel, internal-expanding-type, parking brakes capable of holding the pumping unit in a fixed position during all operating conditions and positions specified herein. The parking brakes shall have only one control.

3.11.1.3 Springs. Springs shall be furnished on the trailer to provide good riding qualities, thereby precluding damage to the mounted equipment. The springs shall be of such construction that not more than 2 inches are added to the frame height for spring deflection.

3.11.1.4 Drawbar assembly. The trailer shall be furnished with a hinged drawbar assembly. The drawbar shall have a lunette eye conforming to Drawing 7041347 with a 3-inch inside diameter and a 1-5/8-inch diameter cross-section.

3.11.1.5 Lifting and tiedown fittings. The trailer shall be equipped with four or more lifting and tiedown fittings in accordance with Drawing 48B7385, 48B7796, or 49B6564. The lifting and tiedown fittings shall be capable of withstanding the loadings specified in MIL-A-8421.

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3.11.1.6 Road clearance. The fully loaded trailer shall provide at least 8 inch road clearance. The tread width shall be 49 inches.

3.11.2 Control panel. A control panel shall be provided for mounting all meters, valves, indicators, gages, switches, and controls furnished with the pumping unit. The control panel shall be constructed of No. 12(0.1046) U.S. gage, or thicker, sheet metal supported by a welded-steel, self-supporting framework. The control panel frame shall be bolted or welded to the trailerbed and positioned so that the wheels, tow bars, et cetera, will not interfere with the operator. The control panel shall be located at sufficient height above the ground and inclined at a sufficient angle (10 degrees minimum) to facilitate observation and operation of all items by an operator approximately 5 ft 10 inches tall standing erect on the ground. Instruments subject to errors from vibration shall be shock mounted.

3.11.2.1 Engine instruments and controls. All engine instruments and controls shall be grouped together on one section of the control panel.

3.11.3 Housing. The pumping unit components and control panel shall be completely enclosed and protected by a weather-resistant housing. The housing shall consist of a welded steel framework covered with No. 16(0.0598) U.S. gage, or heavier, sheet steel with sheet-metal doors and adjustable louvers, as required. The housing shall be capable of supporting a 200-pound (lb) weight distributed evenly over a 4-1/2 by 5 inch area and applied in any location on the top of the pumping unit. The housing shall be braced to prevent distortion or excessive vibration during handling and operation of the pumping unit, or handling of the housing while removed from the pumping unit. The pumping unit shall be designed so that water or hydraulic fluid spilled any place within the housing will drain to the exterior.

3.11.3.1 The doors and louvers in the housing shall be free from binding and easy to operate with one hand. Provisions shall be incorporated to secure the doors rigidly in the full-open or full-closed position. The end panel in the engine compartment shall be replaceable to interchange with the end panel specified in 3.12.2.

3.11.3.2 The housing shall be easily removable by disconnecting fasteners and lifting or sliding the entire housing off the remainder of the pumping unit.

3.11.4 Compartment. A handbook compartment 2 by 9 by 12 inches, or larger, shall be provided and permanently attached within the housing.

3.12 Winterization kit. The AF/M27M-1 pumping unit shall be provided with a winterization kit to convert the pumping unit for operation in ambient temperatures of -65°F.

3.12.1 The winterization kit shall include the following items:

* 3.12.1.1 Hose sets. Three complete hose sets as specified in 3.9.13 through 3.9.13j shall be provided.

* 3.12.1.2 Heater connection. Provisions shall be made for connecting a type H-1 heater conforming to MIL-H-4607, or equivalent ground heater, to the heater duct specified in 3.12.1.3. The heater connection shall be in accordance with Drawing 50C24033 and shall be easily accessible for connecting the

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heater when the housing specified in 3.11.3 is installed. A protective cover shall be provided to prevent the entrance of foreign material through the heater connection when the heater is disconnected.

3.12.1.3 Heater ducts, deflectors, and enclosures shall be provided, as required, to direct the heated air to sections of the engine, battery, et cetera, as necessary to accomplish the -65°F starting specified in Section 4. The duct directing heat to the battery enclosure shall be equipped with an automatic regulating device to prevent overheating of the battery.

3.12.2 The winterization kit shall be easily installed or removed with simple handtools and shall not require welding, cutting, or drilling on the pumping unit or the kit during installation. All connections and fasteners shall be suitable for repeated use. The use of self-tapping, sheet-metal screws shall be prohibited.

3.13 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.14 Dimensions. The length and width of the pumping unit and housing shall not extend beyond the outline of the wheels at any point. The overall height of the pumping unit shall be held to a minimum consistent with good design practices, but shall not exceed 51 inches above the ground with tires fully inflated and without fluid or gasoline.

3.15 Weight. The weight of the pumping unit, including the trailer, shall not exceed 2,500 lbs.

* 3.16 Treatment and painting. Treatment and painting shall be in accordance with MIL-STD-808. Final film shall be designation DG, type I exposure.

3.17 Marking.

3.17.1 Fuel tank cap. The fuel tank cap shall be stenciled to denote the type of fuel to be used.

3.17.2 All items mounted on the control panel shall be identified as to use and cautions by permanently etched or engraved metal or plastic nameplates securely attached to the panel.

3.17.3 A 4- by 6-inch, engraved, permanent-type diagram plate arranged similar to Figure 1 to show the hydraulic circuit of the pumping unit shall be securely fastened to the center of the instrument panel.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special markings shall appear on the nameplate:

TOWING SPEED 20 MPH
GROSS WEIGHT
GASOLINE ENGINE DRIVEN

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3.19 Workmanship.

3.19.1 *General*. The pumping unit, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, and bolts; et cetera.

3.19.2 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.3 Cleaning. The pumping unit shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips; and other foreign material shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of the pumping unit shall be classified as follows:

a. Preproduction testing (4.3)

b. Quality conformance tests (4.4)

* 4.3 Preproduction testing.

* 4.3.1 Preproduction test sample. The preproduction test sample shall consist of one model representative of the production equipment. It shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

* 4.3.2 Preproduction test. Preproduction tests shall consist of all the tests described under 4.6, except 4.6.2.1 and 4.6.3.

4.3.2.1 Initial inspection. The pumping unit submitted for preproduction tests shall be completely inspected and all gages, meters, and instruments laboratory checked against certified calibrating equipment. Failure to meet the accuracy requirements specified in Section 3, for any item, shall be cause for rejection. All requirements of Section 3 which may be determined by measurements shall be checked for compliance and failure to meet the requirements of Section 3 shall be cause for rejection. Following this inspection, the pumping unit shall be reassembled and subjected to the remainder of the preproduction tests.

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* 4.4 Quality conformance tests. The quality conformance tests shall consist of the following:

a. Individual tests

b. Periodic sampling tests

4.4.1 Individual tests. Each pumping unit shall be subjected to the following tests as described under 4.4:

a. Examination of product

b. Operational check

* 4.4.2 Periodic sampling tests. One pump unit shall be selected from each 10 or fraction thereof on the contract or order and subjected to the semi-endurance and cycling test specified in 4.6.3. The test sample shall be selected at random by the Government inspector from production units which have successfully completed the individual tests.

* 4.4.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later production shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and action to preclude recurrence. After correction, all the tests shall be repeated.

4.5 Test conditions.

4.5.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at standard atmospheric pressure (approximately 29.92 in mercury) and at room temperature (approximately 73°F).

4.5.2 Test time. Unless otherwise specified, any test shorter than 1 hour shall not be credited as test time.

4.5.3 Apparatus and instruments. Apparatus and instruments used in conjunction with testing specified herein shall be of laboratory precision type and shall be carefully checked to ascertain that they have been installed in a satisfactory manner for durability and accurate operation. They shall be calibrated at intervals properly spaced to assure laboratory efficiency.

4.5.4 Fluid flow. The calibration and performance checks of the displaced fluid shall be determined by means of a calibrated flowmeter with certified accuracy of 2 percent of reading or better.

4.5.5 Hydraulic fluid. Unless otherwise specified (see 6.2), the hydraulic fluid used in the pumping unit shall conform to MIL-H-5606.

4.6 Test methods.

4.6.1 Examination of product. The pumping unit shall be inspected to determine compliance with the requirements specified herein with respect to parts, materials, workmanship, dimensions, and markings.

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* 4.6.2 Operational check. The constant flow operation of each pumping unit shall be checked for 10 min at 800, 2000, and 3000 pounds per square inch at a rate of 4 gal/min. The test circuit for the constant flow tests shall be as specified in 4.6.5. During and after the constant flow tests, the pumping unit shall be checked for fluid leakage, overheating, excessive vibration, and general satisfactory performance. The reservoir fluid level gage shall be checked for accuracy and shall indicate the actual amount of fluid in the reservoir within ±2 gal throughout its range. The satisfactory operation of the high-pressure relief valve shall be checked at 3,000 and 3,500 pounds per square inch gage.

4.6.2.1 The functional operation of the pumping unit shall be checked by conducting five cycles of the cycling tests specified in 4.6.13.

4.6.3 Semi-endurance and cycling. The pumping unit shall be subjected to 30 hours of the endurance test time specified in 4.6.5 and 20 cycles of the cycling tests specified in 4.6.13. Upon completion of this test, the pumping unit shall satisfactorily pass the test specified in 4.6.2.

4.6.4 Environmental tests. The pumping unit shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810. Upon completion of each test, the pumping unit shall be subjected to each of the tests specified in 4.6.2. Failure of the pumping unit to operate as required during or after each exposure, or failure of any item or assembly due to deterioration from exposure in this test, or formation of any undesirable condition which cannot be remedied by routine maintenance shall be cause for rejection.

4.6.4.1 Low temperature tests.

4.6.4.1.1 AF/M27M-1 pumping unit. Lubricants shall be replaced by oils or grease suitable for -65°F operation as required. The fluid in the hydraulic system and reservoir shall be completely drained and replaced by a mixture by volume of 75 percent hydraulic fluid in accordance with MIL-H-5606 and 25 percent aliphatic naphtha in accordance with TT-N-95. The pumping unit shall be connected to the test circuit specified in 4.6.5 and the pumping unit and test circuit shall then be stabilized at -65°F in accordance with procedure I. A Type H-1 heater conforming to MIL-H-4607, or an equivalent ground heater, shall be connected to the pumping unit and heat applied. The pumping unit shall start within 30 minutes after heat is first applied and shall attain rated output within 15 minutes after starting. The heater shall be shut down and the pumping unit shall continue to operate satisfactorily at full capacity for a period of 1 hr with the ambient temperature in the chamber at -65°F. All valves, gages, doors, et cetera, shall be checked for satisfactory operation at -65°F. The pumping unit shall then be removed from the chamber and the winterization kit removed and replaced by the original equipment and diluted fluid replaced by hydraulic fluid in accordance with MIL-H-5606 for the remainder of the tests.

4.6.4.1.2 AF/M27M-2 pumping unit. The pumping unit shall be subjected to low temperature in accordance with procedure I. After 72 hours of exposure at -65°F, the ambient temperature in the coldroom shall be increased to +30°F or above, and each test specified in 4.6.2 shall be conducted. The temperature of the coldroom shall be lowered to -20°F. After 72 hours of exposure at -20°F, starting procedures shall be accomplished. Failure of the pumping unit to start immediately without aid of external heat shall be cause for rejection. The pumping unit shall attain rated output of 4 gal/min and 3,000 pounds per

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square inch within 15 minutes after starting, and shall operate satisfactorily for 30 minutes at 4 gal/min and 3,000 pounds per square inch in the -20°F ambient temperature. All gages, meters, valves, accessories, louvers, et cetera, shall be operated and shall perform satisfactorily in the -20°F ambient temperature.

4.6.4.2 High temperature. The pumping unit shall be subjected to high temperature in accordance with procedure I. Upon completion of the exposure, the ambient temperature shall be lowered to 130°F, and after stabilizing at 130°F, the tests specified in 4.6.2 shall be conducted.

4.6.4.3 Salt fog. The pumping unit shall be subjected to salt fog in accordance with the procedure specified.

4.6.4.4 Rain. The pumping unit shall be subjected to rain in accordance with the procedure specified. The pumping unit shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for 2 hours and while still in the rain test, the pumping unit shall be started and operated at 4 gal/min and 3,000 pounds per square inch for a period of 30 minutes. Failure of the pumping unit to start or operate satisfactorily during the rain test shall be cause for rejection. Upon completion of the rain test, all surfaces which might collect moisture shall be inspected to determine adequacy of drainage facilities and any collection of water in excess of 1/8 in or which could be detrimental to the performance of the pumping unit shall be cause for rejection.

4.6.4.5 Fungus resistance. The pumping unit shall be subjected to fungus in accordance with the procedure specified.

4.6.4.6 Humidity. The pumping unit shall be subjected to humidity in accordance with the procedure I.

4.6.5 Endurance. The output flow from the pumping unit shall be connected through the nine outlets of the hose sets to an external test circuit. The external test circuit shall consist of a manifold, variable restrictor, cooler, flowmeter, and a return line to the reservoir of the pumping unit. The pumping unit shall be equipped with nine connectors in accordance with Drawing 43A13906 for connection to the hose sets and one outlet connected to the restrictor. The restrictor shall be adjustable to maintain 3,000 pounds per square inch gage pressure and a total of 4 gal/min flow from the pumping unit. The cooler shall be of adequate capacity to maintain fluid temperature entering the reservoir at 130°F, or less. The flowmeter shall have a certified accuracy of 2 percent of reading, or less, throughout a range of 3 to 5 gal/min. The output from the flowmeter shall be returned to the reservoir of the pumping unit. The pumping unit shall then be operated through the test circuit at 4 gal/min and 3,000 pounds per square inch gage for a total of 150 hours. The pumping unit shall operate satisfactorily throughout the 150-hour endurance test without adjustments, repairs, or replacement of any item. The pumping unit shall then be subjected to and shall satisfactorily pass the performance tests specified in 4.6.2.

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4.6.6 Gasoline engine tests. During the endurance tests, the following readings shall be recorded every hour and the data collected shall be evaluated. Data shall show that engine operation is normal as recommended by the engine manufacturer and that no unsatisfactory conditions exist.

a. Engine revolutions per minute

b. Fuel consumption

c. Oil consumption

d. Hourmeter

e. Oil pressure

f. Cylinder head temperature (read for all cylinders)

g. Ammeter

h. Any other engine instrument provided

4.6.6.1 The horsepower of the gasoline engine at rated revolutions per minute shall be determined by a driven dynamometer and calculations made to determine available horsepower at 6,000 ft altitude pressure and 130°F. The horsepower required by the hydraulic pump while pumping fluid in accordance with MIL-H-5606 at -20°F and at a rate of 4 gal/min at 3,000 pounds per square inch gage shall be determined. The horsepower of the engine at 6,000 ft altitude and 130°F ambient times 0.75 shall be equal to or greater than the above pump requirement.

4.6.6.2 The pumping unit shall be started and operated at 4 gal/min and 3,000 pounds per square inch until oil and cylinder head temperatures are stabilized. Readings in revolutions per minute shall then be taken at the maximum and minimum operating speeds provided by the adjustable range governor and with full-, one-half, and one-quarter load suddenly applied and then suddenly removed to check governor regulation. The governor shall maintain revolutions per minute within 5 percent of setting under all load conditions.

4.6.6.3 Fuel consumption readings taken during the endurance run shall be averaged and gallons per hour average shall equal or be less than one-fourth of the available capacity of the fuel tank.

4.6.6.4 The engine shall be examined to determine that satisfactory provisions are incorporated in the fuel system and elsewhere to permit starting and operation at altitudes up to 6,000 ft and to verify the evidence submitted as specified in 3.6.1.

4.6.7 Electromagnetic interference. The pumping unit shall be tested for electromagnetic interference in accordance with MIL-STD-461 (see 3.8.3).

4.6.8 Noise level. The noise level of the pumping unit while operating at full capacity shall be measured 10 ft from the perimeter of the test specimen. Measurements shall be taken out-of-doors on a hard surface from at least eight different directions, 45 degrees apart and in a horizontal plane parallel to the trailerbed and passing through the centerline of the high-pressure pump driveshaft. Noise values shall not exceed the limits of the various frequency bands as specified below:

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OCTAVE BAND IN CPS	OCTAVE BAND LEVEL IN DECIBELS (REF. 0.0002 MICROBAR)
20 - 75	108
75 - 150	98
150 - 300	90
300 - 600	83
600 - 1,200	78
1,200 - 2,400	75
2,400 - 4,800	72
4,800 - 10,000	71

4.6.9 Operation in the tilted position. The pumping unit shall rest on a platform and shall be kept from rolling by the parking brakes. The platform supporting the pumping unit shall be tilted in eight different directions, 45 degrees apart in an 8-1/2 degree incline with the true horizontal plane. The pumping unit shall be started and operated at 4 gal/min and 3,000 pounds per square inch for 30 minutes in each position. The pumping unit shall perform satisfactorily in every respect and the parking brakes shall prevent the pumping unit from moving on the platform. While tilted in each of the eight positions, 38 gal of fluid shall be pumped from the reservoir to check the capacity specified in 3.9.8.

4.6.10 Static load. A 200-lb weight shall be applied slowly and uniformly to a 4.5 by 5 inch area in at least six random positions on top of the pumping unit housing, including a reading in the center of each panel. The pumping unit housing shall not be permanently deformed in any way by the application of the weight.

4.6.11 Mobility. The pumping unit shall be subjected to and shall pass the tests specified in MIL-M-8090 for type II mobility. Upon completion of the mobility tests, the pumping unit shall be subjected to the tests specified in 4.6.2.

4.6.12 Flow control valves. The flow control valves shall be tested by connecting two of the feeder hoses, which attach to the hose reels, to both sides of a double-acting hydraulic piston. The other side of one of the control valves shall be connected to the output side of the selector valve and the other side of the remaining control valve shall be connected to the return system. With the selector valve directing pressure to one side of the double-acting piston (with the control valve on the pressure side full open) the control valve in the return line shall allow the piston to move to any position and be capable of stopping the piston at any time without creep. The above procedure shall be repeated to test the other two control valves. The working pressure for this test shall be 3,500 pounds per square inch. Failure of the valves to operate as specified in 3.9.6 or to satisfactorily manipulate the flow to the hydraulic cylinder or to stop the piston in any position without creep shall be cause for rejection.

* 4.6.13 Cycling tests. The pumping unit shall be connected through the nine connectors of the hose sets to nine 2-1/2-gal accumulators Greer Hydraulics Inc. model No. 30A-2-1-1/2A, or equal, to simulate the operation of nine aircraft jacks. Two accumulators in each hose set shall be charged with air to 1,000 pounds per square inch gage and the two remaining in each hose set shall be charged with air to 500 pounds per square inch gage. The accumulator in the single hose shall be charged to 1,000 pounds per square inch gage. The following cycle shall then be repeated one hundred times:

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a. Move selector valve to the up position and gradually build up fluid pressure in the nine accumulators to 3,000 pounds per square inch gage by slowly opening the control valves.

b. Two minutes - Hold 3,000 pounds per square inch gage pressure in the accumulators, control valves full open.

c. Ten minutes - Move the selector valve to the hold position and maintain for 10 minutes. During this period, not more than 1 gal of fluid shall leak back to the reservoir through the selector valve.

d. Move the selector valve rapidly to the down position and hold until all the hydraulic fluid has been discharged from the accumulators. During this period, the suction relief valve shall permit adequate flow to the pump to prevent pump cavitation.

e. Close control valves and move selector valve to the hold position to complete the cycle.

* 4.6.13.1 Suction relief valve. During the cycling tests, the suction relief valve shall be adjusted to 2, 6, and 8 inches of mercury vacuum (2 inches of mercury for 25 cycles, 6 inches mercury for 50 cycles), and satisfactory operation checked by reading the vacuum gage during the last 30 seconds of step 4.6.13d, while the selector valve is in the down position and flow from the accumulators has stopped. The suction relief valve shall provide adequate flow to the pump to prevent pump cavitation.

4.6.13.2 The one hundred cycles of the cycling tests shall be completed without the necessity of replacing parts or performing maintenance. The suction filter and all components shall perform satisfactorily during the test and inspection shall show no fluid leaks or damage to the filter elements.

4.6.14 Flow regulators. The nine flow regulators specified in 3.9.13h shall be tested at 800, 1000, 2000 and 3000 pounds per square inch and each regulator shall restrict return flow to 0.7 ±0.05 gal/min at each pressure specified.

4.6.15 Serviceability. The pumping unit shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular note shall be made with regard to provisions to prevent accumulation of dirt, snow, ice, et cetera, which may hinder servicing and operation.

4.6.16 Final inspection. The pumping unit shall be disassembled as required and a thorough examination made of all parts to disclose failure, distortion, excessive wear, or other unsatisfactory conditions. Results shall be compared with similar results determined during initial inspection. Pumping units not conforming to the requirements specified herein shall be rejected.

4.6.17 Air-transportability. The pumping unit shall be subjected to the acceleration loadings specified in MIL-A-8421.

* 4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

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5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A (Mobile), Level B (Mobile) or Level C (Mobile), as specified (see 6.2), in accordance with MIL-STD-281.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Levels A and B. Each pumping unit, preserved and packaged as specified in 5.1 and fitted (covered) with a shroud fabricated and marked in accordance with MS27766, shall be shipped as a mobile unit.

* 5.2.2 Level C. Each pumping unit preserved and packaged as specified in 5.1 shall be shipped as a mobile unit.

* 5.3 Marking. In addition to specified special marking the pumping units shall be marked in accordance with the applicable requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The AF/M27M-1 and AF/M27M-2 pumping units covered by this specification are intended for use in simultaneous operation of nine aircraft hydraulic jacks.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type pumping unit required. (See 1.1)

c. Hydraulic fluid if other than as specified. (See 4.5.5)

d. Levels of preservation and packaging, and packing required. (See Section 5)

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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